

GreenGuard CO2 Trap Timer

Battery-powered inline CO2 solenoid timer | 4-digit display | Simple 2-button time set | Rev 2.0 | June 2026
ON: 05:30 | OFF: 23:30 | Universal — compatible with all CO2 mosquito trap systems

PRODUCT OVERVIEW

The GreenGuard CO2 Timer automates CO2 release on a fixed daily schedule, cutting overnight consumption while targeting peak mosquito activity hours. Universal replacement for the Biogents BG-CO2 Timer (\$99.99, chronically out of stock). Compatible with all CO2 mosquito trap systems. Retail: \$79.99.

OPERATING SCHEDULE

CO2 ON	5:30 AM daily
CO2 OFF	11:30 PM daily
Off Window	6 hrs overnight
CO2 Savings	~25% vs. continuous

TIME SETTING — 2 BUTTON + DISPLAY

TO SET TIME:	
1.	Hold SET button 3 seconds — display flashes
2.	Press UP to set hour (00-23)
3.	Press SET to confirm hour
4.	Press UP to set minutes
5.	Hold SET 2 seconds to save

ELECTRICAL SPECIFICATIONS

MCU	ATtiny85-20PU
RTC	DS3231 (+-2ppm accuracy)
Display	TM1637 4-digit 7-segment
Power	9V alkaline battery
Battery Life	9-12 months typical
Sleep Current	~5 uA (deep sleep)
Solenoid Drive	12V NC via IRL540N MOSFET
Fail Safe	CO2 OFF on power loss
PCB	60x40mm, 2-layer FR4

MECHANICAL SPECIFICATIONS

Enclosure	IP65 ABS 100x68x50mm
CO2 Fittings	1/4" NPT to 6mm barb (brass)
Cable Entry	2x PG7 IP65 cable glands
PCB Mount	M3 brass standoffs 10mm
Display	TM1637 module, lid cutout 33x14mm
Buttons	2x 6mm tactile, front face

ASSEMBLY NOTES

- ATtiny85 must be ICSP-flashed via J3 BEFORE board assembly. File: `co2_timer_v2.ino`
- SMT parts (U2, U3, F1, R1-R6, C2/C3) assembled by JLCPCB. Through-hole parts hand-soldered.
- DS3231 VBAT pin ties to 3.3V rail — no coin cell required.

BILL OF MATERIALS

Ref	Component	Part / Source	Qty
U1	ATtiny85-20PU MCU	LCSC C89852	1
U2	DS3231SN RTC	LCSC C9868	1
U3	AMS1117-3.3 LDO	LCSC C6186	1
Q1	IRL540NPBF N-MOSFET	LCSC C60330	1
D1-D3	1N4007 Diode (x3)	LCSC C727110	3
SW1	UP Button 6x6mm	LCSC C318884	1
SW2	SET Button 6x6mm	LCSC C318884	1
F1	Polyfuse 1A 1812 SMD	LCSC C369150	1
R1-R6	Resistors 10K/4K7/100R 0805	LCSC (0805 SMD)	6
C1	100uF 16V Electrolytic	LCSC C62923 (THT)	1
C2/C3	100nF Ceramic (x2)	LCSC C49678 (0805 SMD)	2
J1	Screw Term. 2P 5.08mm	LCSC C8465 — Battery in	1
J2	Screw Term. 2P 5.08mm	LCSC C8465 — Solenoid out	1
J3	ICSP Header 2x3 2.54mm	LCSC C124378	1
—	TM1637 Display Module	Amazon / LCSC — 4-digit	1
—	Solenoid 12V NC Viton 1/4"	US Solid (Amazon)	1
—	IP65 ABS Enclosure	Zulkit 100x68x50mm	1
—	1/4" NPT to 6mm Barb (x2)	Amazon	2
—	PG7 Cable Glands (x2)	Amazon	2
—	9V Battery Snap	Amazon	1
—	M3 Standoffs 10mm (x4)	Amazon	4

FILES PROVIDED

co2_timer_gerbbers_JLPCPB.zip	PCB Gerber files
JLPCPB_BOM.csv	BOM with LCSC part numbers
co2_timer-CPL-JLPCPB.csv	Component placement (CPL)
co2_timer.kicad_pcb	KiCad PCB layout file
co2_timer_v2.ino	ATtiny85 firmware source

QUOTE REQUEST

Initial Qty	10 units
Future Qty	100 / 500 units
Assembly	Turnkey preferred
Programming	ATtiny85 ICSP flash at assembly
Deliverable	Fully assembled box build
Contact	admin@greenguard-usa.com
Phone	512-560-4129

UNIT COST & MARGIN